# BANNARI AMMAN INSTITUTE OF TECHNOLOGY
# &
# INSTITUTIONS INNOVATION COUNCIL

# Social Relevant Project

# Second Year

## BASIC STUDENT DETAILS


**Student Name** : **ARUN KUMAR S**

**Registration Number** : **7376242AD122**

**Department** : **AI & DS**

**Mail id** : **arunkumars.ad24@bitsathy.ac.in**

**Domain** : **Software**


## BASIC PROJECT DETAILS

**Project Title** : Farmers Disease Diagnostic Reporting Portal – Mobile Based Portal

**Problem Bank Reference / ID** : Agriculture – SRP – Crop Health & Disease Monitoring

**Domain** : Agriculture Technology (Agri Tech)

**Application Area** : Crop Disease Identification, Farmer Support System

**SDG Number(s) & Title(s)** : SDG 2 – Zero Hunger & SDG 9 – Industry, Innovation and Infrastructure

**SDG Justification** : Early identification of crop diseases helps farmers reduce yield loss and improve agricultural productivity. The proposed solution supports sustainable farming practices and improves

food security by providing timely disease diagnosis and guidance to farmers.

# PHASE 1: PROBLEM IDENTIFICATION

1.1 **Problem Statement:**

Farmers in rural and semi-rural areas face significant challenges in identifying crop diseases at an early stage. Due to limited access to agricultural experts, lack of awareness, and dependency on traditional practices, farmers often fail to diagnose plant diseases correctly. This results in delayed treatment, spreading of diseases across fields, and severe crop yield loss.

The problem is more critical for small and marginal farmers who depend entirely on agriculture for their livelihood. Incorrect diagnosis leads to the use of wrong pesticides, increased cost of cultivation, environmental harm, and reduced income. A reliable, easy-to-use mobile-based disease diagnostic reporting system is required to support farmers with timely and accurate information.

**1.2 Social Relevance:**
Tick applicable:

☑ Health & Safety

☐ Time Loss

☑ Financial Loss

☑ Environmental Impact

☑ Quality of Life

**Justification :**
Crop diseases directly affect farmers' income and food production. Delayed or incorrect treatment leads to financial loss and mental stress among farmers. Excessive pesticide use harms soil health and the environment, impacting long-term agricultural sustainability.

**1.3 Location & Context:**

**Location Type:**

☐ Urban

☐ Semi-Urban

☑ Rural

**Specific Location:**

Sathyamangalam.

Narapanatti(vill) ,Krishnagiri(Dt)

**Scale of Impact:**

☐ Individual

☑ Community

☑ Regional

**1.4 Affected person details:**

| **Problem facer Category** | **Level of impact** | **Importance to address this issue** | **Data available (if any)** |
|---|---|---|---|
| Farmers | Agriculture | Direct impact on livelihood | Crop loss data, yield reports |
| Farm Workers | Agriculture | Affects employment | Observational data |
| Agricultural Vendors | service | Business dependency | Sales records |

| | | | |
|---|---|---|---|
| | | | |

**1.5 Stakeholder Identification:**

| Stakeholder Category | Domain mapping | Details | Any website identified |
|---|---|---|---|
| Farmers | Agriculture | Primary beneficiaries | http://www.tnagrisnet.tn.gov.in |
| Agricultural Officers | Government | Advisory & validation | http://www.agricoop.gov.in |
| Agri Experts | Research | Disease diagnosis | http://www.icar.org.in |
| Mobile App Users | Technology | Platform users | — |
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## PHASE 2: FIELD STUDY & BENEFICIARY SURVEY

**2.1 Purpose of Field Study:**
Tick applicable:

☑Understand real challenges

☑Validate problem

☑Study current practices

☑Identify limitations

☑Collect requirement data

**2.2 Field Visit Details:**

| **Item** | **Details** |
|---|---|
| LOCATION | Rural farming areas |
| DATE | January 18 ,2026 |
| METHOD | Direct interaction |
| PARTICIPANTS | Farmers |
| | |
| | |
| | |

**2.3 Beneficiary Profile (Minimum 5–10):**

| **ID** | **Age Group** | **Occupation** | **Location Type** | **Contact (Name, designation, organization, phone number)** | **Signature** |
|---|---|---|---|---|---|

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| | | | | | |
|---|---|---|---|---|---|
| B1 | 50-55 | Farmer | Rural | Mr. Ramesh 9047867611 | Ramesh |
| B2 | 48 | Farmer | Rural | Mr.Srinivasan 7402419470 | srinivasan |
| B3 | 35 | Farmer | Rural | Mr.Mohan 7373858264 | Mohan |
| B4 | 42 | Homemaker, Farmer | Rural | Mrs. Kalavathi 7695956914 | Kalavathi |
| B5 | 28 | Farmer | Rural | Mr.sriram 6380216113 | Sriram |
| B6 | 65 | Farmer | Rural | Mrs.Gowramma 9385620483 | Gowramma |
| | | | | | |
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## 2.4 Interaction Survey Questions (Survey questions has to taken for each beneficiary separately)

*(Tick ONE option unless mentioned otherwise)*

# Beneficiary details: Person 1/2/3/4/5/6/7/8/9/10 (As per table in 2.3)

### A. Problem Occurrence & Impact

1. **How often do you face this problem?**

   ☐Daily ☐Weekly ☑Monthly ☐Rarely

2. **At what time does the problem mostly occur?**

   ☐Morning ☐Afternoon ☐Evening ☑Night ☐Anytime

3. **How long does the problem usually last?**

   ☐< 10 minutes ☐10–30 minutes ☐30–60 minutes ☑> 1 hour

4. **Severity of the problem**

   ☐Very High ☑High ☐Moderate ☐Low

5. **Primary impact of the problem** *(Select one)*

   ☐Safety risk

   ☑Financial loss

   ☐Health issue

☐Time loss

☐Inconvenience only

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## B. Current Practice & Limitations

6. **Current method used to handle the problem**
   ☐Manual method
   ☐Existing tool / system
   ☑External help (person/service)
   ☐No solution available
7. **Effectiveness of the current method**
   ☐Very effective
   ☐Moderately effective
   ☑Slightly effective
   ☐Not effective
8. **Main difficulty in the current method** *(Select one)*
   ☐High cost
   ☐Time consuming
   ☐Difficult to operate
   ☑Not reliable
   ☐Not always available
9. **Frequency of failure in current method**
   ☑Very frequent
   ☐Sometimes
   ☐Rare
   ☐Never

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## C. Technology & Infrastructure Constraints

10. **Comfort level in using technology**
    - ☐ Very comfortable
    - ☑ Can use with guidance
    - ☐ Difficult to use
    - ☐ Cannot use at all
11. **Electricity availability at location**
    - ☐ Always available
    - ☑ Mostly available
    - ☐ Rarely available
    - ☐ Not available
12. **Internet availability**
    - ☐ Good
    - ☑ Limited
    - ☐ Poor
    - ☐ Not available
13. **Device availability** *(Select applicable)*
    - ☑ Smartphone
    - ☐ Basic mobile
    - ☐ Computer
    - ☐ No device

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## D. Cost Sensitivity & Acceptance

14. **Acceptable cost level**
    - ☐ No cost
    - ☑ Low cost
    - ☐ Medium cost
    - ☐ High cost acceptable
15. **Preference for solution type**
    - ☑ Software based
    - ☐ Hardware based

☐Process / method based

☐Combination

16. **Willingness to adopt a new solution**

☐Yes, immediately

☑Yes, after testing

☐Maybe

☐No

17. **Willingness to recommend the solution to others**

☑Yes

☐No

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## Geotag photo (Attach the image here)

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# 2.5 Field Observations

*(Tick all that apply based on direct observation)*

## A. Operational Issues

☐Workarounds observed

☐Repeated manual effort

☑Process delays

☑Dependency on others

## B. Safety & Risk Factors

☐Safety hazards present

☐Risk of injury

☑Health-related risk

☐No safety issues observed

## C. Resource & Efficiency Issues

☐Resource wastage (time/material/energy)

☑High operational cost

☑Inefficient workflow

☐Redundant steps

## D. User Behaviour Patterns

☐Users ignore the problem

☐Users adapt temporary fixes

☑Users depend on external help

☑Users are open to improvement

**Common pattern identified:**

Farmers rely on guesswork or external agents for disease identification and lack a quick,

, reliable diagnostic solution.

**Contradictions or unexpected insights identified:**

Despite owning smartphones, farmers rarely use digital tools for agriculture.

**Observation Summary (MANDATORY – 5 to 6 lines only)**

*(Based strictly on what was **seen**, not assumptions)*

**Most farmers experience crop disease issues regularly and depend on external support for diagnosis. The existing methods are time-consuming and unreliable. Farmers own smartphones but lack agriculture-specific applications. Internet availability is limited, but users are willing to adopt new technology if it is simple and affordable.**

# PHASE 3: REQUIREMENT STUDY

## 3.1 Functional Requirements

*(Derived strictly from Phase-2 survey responses and field observations)*

| Req ID | Requirement Description | Derived From (Survey Q.No / Observation) | Priority |
|---|---|---|---|
| FR-01 | System shall allow farmers to upload crop images | Q.No 6 | High |
| FR-02 | System shall identify crop diseases | Q.No 4 | High |
| FR-03 | System shall provide treatment suggestions | Q.No 5 | High |
| FR-04 | System shall support local language | Q.No 10 | Medium |
| FR-05 | System shall store diagnostic reports | Observation | Low |
| | | | |
| | | | |
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| | | | |

 *Example:*
"System shall provide alerts" → Derived from Q2 (High severity)

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## 3.2 Non-Functional Requirements (Select All Applicable)

### A. Usability & Accessibility

☑Easy to learn
☑Minimal user input
☑Local language support
☐Suitable for low literacy users

### B. Cost & Affordability

☐No recurring cost
☑Low initial cost
☑Affordable for mass users

### C. Performance & Reliability

☑Fast response
☑Accurate output
☐Works consistently under normal conditions

### D. Safety & Compliance

☐Physically safe to use
☑Data privacy protected
☐Complies with legal norms

### E. Maintenance & Sustainability

☑Low maintenance
☐Easily repairable
☐Environment friendly

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## 3.3 Constraints Analysis

*(Derived from field conditions – tick all applicable)*

### A. Infrastructure Constraints

☐Limited power supply
☑Power outages
☑Limited internet connectivity
☐No internet availability

### B. User-Related Constraints

☑Low technical literacy
☐Resistance to change
☑Dependence on external help

### C. Environmental Constraints

☐Outdoor usage
☐Dust / heat / humidity exposure
☑Seasonal variations

### D. Legal & Safety Constraints

☐Safety regulations
☐Government policies
☑Ethical constraints

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## 3.4 Mandatory vs Optional Requirements

*(Based on feasibility and beneficiary priority)*

| Requirement ID | Requirement Description | Type | Reason |
|---|---|---|---|
| FR-01 | Image upload | Mandatory | Core functionality |
| FR-02 | Disease diagnosis | Mandatory | Critical need |
| FR-03 | Treatment guidance | Mandatory | Farmer requirement |
| FR-04 | Local language | Optional | Improves usability |

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## 3.5 Requirement Validation Summary

*(Tick based – no open writing)*

☑Requirements approved by beneficiaries
☑Requirements feasible under constraints
☑Requirements aligned with social relevance
☑Requirements suitable for next semester execution

**Signature of the Student**

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## BASIC STUDENT DETAILS


**Student Name** : ARUN KUMAR S

**Registration Number** :7376242AD122

**Department** :ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

**Mail id** : arunkumars.ad24@bitsathy.ac.in


**Domain** : Software

## BASIC PROJECT DETAILS

**Project Title** : FARMERS DISEASE DIAGNOSTIOC REPORTING PORTAL-MOBILE BASED

**Problem Bank Reference / ID** :S0415

**Domain** : SOFTWARE

**Application Area** : Agricultural Technology (AgriTech), Crop Disease Diagnosis, Farmer Support

**SDG Number(s) & Title(s)** : SDG 9 – Industry, Innovation and Infrastructure

**SDG Justification** :

The proposed mobile-based disease diagnostic reporting portal supports sustainable agriculture by enabling early detection of crop diseases. Early diagnosis reduces crop loss, improves farmer income, and enhances

food production efficiency. By promoting digital innovation in agriculture, the solution contributes to food security (SDG 2), technological advancement in rural sectors (SDG 9), and responsible pesticide usage (SDG 12).

# PHASE 4: JOURNEY MAPPING

**4.1 Objective of Journey Mapping**

☑ Understand complete user experience

☑ Identify exact pain points

☑ Detect hidden inefficiencies

☑ Identify intervention opportunities

☑ Validate requirement assumptions

**4.2 User Journey Map (Step-by-Step Flow)**

| Step No | User Action | User Thought | User Feeling | Problem Occurrence (Y/N) | Pain Point Description |
|---|---|---|---|---|---|
| 1 | Farmer observes unusual spots on leaves | "Is this normal or disease? " | Confused | Y | Cannot identify disease correctly |
| 2 | Asks nearby farmers for advice | "Maybe they know better" | Uncertain | Y | Information not reliable |
| 3 | Visits pesticide shop | I need quick solution | Worried | Y | Seller suggests general pesticide |
| 4 | Buys pesticide | "Hope this works" | Financial stress | Y | No scientific diagnosis |

| | | | | | |
|---|---|---|---|---|---|
| 5 | Applies pesticide | "Let me try this" | Hopeful | Y | Wrong chemical sometimes used |
| 6 | Waits for result | "Did I do the right thing" | Anxious | Y | Delay in proper treatment |

**4.3 Pain Point Identification**

**Major Pain Points Identified:**

1. Lack of accurate disease identification

2. Dependence on unverified external advice

3 .Financial loss due to incorrect pesticide usage

4.Delay in expert consultation

5 .Spread of disease before detection

**Minor Pain Points Identified:**

1. Time wasted traveling to shops.

2.Stress and uncertainty.

3.Lack of record keeping of past diseases.

**4.4 Opportunity Areas for Engineering Intervention**

| Pain Point | Root Cause | Possible Intervention Area |
|---|---|---|
| Wrong diagnosis | No expert access | Image based disease detection system |
| Financial loss | Trial - and - error pesticide use | AI - based recommendation engine |
| Delay in support | Manual consultation | Real- time digital reporting portal |

**4.5 Journey Map Summary (5–6 lines only)**

1. Farmers first detect symptoms visually but lack knowledge to confirm disease.

2. They depend on nearby farmers or pesticide sellers.

3. Diagnosis is mostly guess-based and inconsistent.

4. Wrong treatment leads to financial and crop loss.

5.A digital intervention at early detection stage can prevent major losses.

# PHASE 5: BOUNDARY CONDITIONS & CONSTRAINT ANALYSIS

## 5.1 Boundary Conditions (What CANNOT be changed)

| **Category** | **Boundary Condition Identified** | **Reason** |
|---|---|---|
| User Literacy | Farmers may have low education level | Cannot change educational background |
| Internet Access | Rural network limitations | Infrastructure controlled externally |
| Device Ownership | Not all farmers use high-end phones | Economic limitations. |

| | | |
|---|---|---|
| Seasonal Farming | Disease occurrence seasonal | Natural condition |
| Financial Status | Farmers are cost sensitive. | Income Instability |

**5.2 System Constraints**

**Infrastructure Constraints:**

☑ Limited power supply

☑ Frequent power cuts

☐ Limited internet

☐ No internet

Others :_____

**User Constraints:**

☑Low literacy

☑ Low digital skills

☐ Resistance to change

☑Cost sensitive

Others :_____

**Environmental Constraints:**

☑ Outdoor usage

☑ High temperature

☑Dust / moisture exposure

Others :_____

**Safety & Legal Constraints:**

☐ Government approval required

☐ Safety compliance required

☑ Data privacy concern

Others :_____

### 5.3 Current System Bottlenecks

| Bottleneck | Cause | Impact Level (H/M/L) |
|---|---|---|
| Delayed diagnosis | Manual consultation | High |
| Wrong pesticide usage | No scientific analysis | High |
| No disease history records | Lack of documentation | Medium |

**(H- High, M- Medium, L – Low )**

### 5.4 Feasibility Risk Identification

| Risk | Probability (H/M/L) | Impact (H/M/L) | Mitigation Strategy |
|---|---|---|---|
| Farmers not adopting app | Medium | High | Provide simple UI & training |
| Poor Internet usage | High | Medium | Enable low-data mode |
| Incorrect AI prediction | Medium | High | Validate with agricultural datasets. |

**(H- High, M- Medium, L – Low )**

## 5.5 Constraint Analysis Summary (5 lines only)

1. Rural infrastructure limits high-bandwidth solutions.

2. Farmers require simple, low-cost solutions

3. Environmental conditions require durable digital access.

4.Digital literacy varies among users.

5. Solution must function under limited connectivity.

## PHASE 6: SOLUTION IDEATION (Minimum 5 Concepts Mandatory)

**6.1 Solution Concept – 1**

**Concept Type:**

### ☑Software-Based

**Core Idea:**

Mobile app allowing farmers to upload crop images for AI-based disease detection.

**How It Solves the Problem:**

Provides instant preliminary diagnosis without waiting for external help.

**Advantages:**

- Fast response
- Scalable solution

**Limitations:**

- Requires internet
- AI accuracy dependent on dataset

**Feasibility Level:**

☑ High

☐ Moderate

☐ Low

**6.2 Solution Concept – 2**

**Concept Type:**

☑ **Hybrid**

**Core Idea:**

Mobile app + agricultural expert validation panel.

**How It Solves the Problem:**

AI gives preliminary result; expert confirms before recommendation.

**Advantages:**

• More reliable

• Builds farmer trust

**Limitations:**

• Requires expert availability

• Slower than pure AI

**Feasibility Level:**

☐ High

☑ Moderate

☐ Low

**6.3 Solution Concept – 3**

**Concept Type:**

☑ **Software-Based**

**Core Idea:**

Offline symptom-based questionnaire system.

**How It Solves the Problem:**

Farmers answer guided questions; system suggests possible disease.

**Advantages:**

• Works without internet

• Easy for low-end devices

**Limitations:**

• Less accurate

• Depends on user input

**Feasibility Level:**

☑ High

☐ Moderate

☐ Low

**6.4 Solution Concept – 4**

Concept Type:

☐ Software-Based

☐ Hardware-Based

☑**Process-Based**

☐ Hybrid

**Core Idea:**

Village-level digital disease reporting center.

**How It Solves the Problem:**

Shared device operated by trained volunteer.

**Advantages:**

• Helps non-digital farmers

• Community approach

**Limitations:**

•Needs manpower

• Not fully scalable

**Feasibility Level:**

☐ High

☑ Moderate

☐ Low

**6.5 Solution Concept – 5**

**Concept Type:**

☐ Software-Based

☐ Hardware-Based

☐ Process-Based

☑ **Hybrid**

**Core Idea:**

Mobile app integrated with weather data for disease prediction alerts.

**How It Solves the Problem:**

Predicts disease outbreaks based on seasonal patterns.

**Advantages:**

• Preventive approach

• Reduces future disease spread

**Limitations:**

• Requires data integration

•More complex development

**Feasibility Level:**

☐ High

☑Moderate

☐ Low

**6.6 Comparative Analysis of All Concepts**

| Criteria | Concept 1 | Concept 2 | Concept 3 | Concept 4 | Concept 5 |
|---|---|---|---|---|---|
| Cost | Low | Medium | Low | Medium | Medium |
| Accuracy | High | Very High | Medium | Medium | High |
| Scalability | High | Medium | High | Low | Medium |
| Implementation Complexity | Medium | High | Low | Medium | High |

**6.7 Preliminary Shortlisting**
**Shortlisted Concept(s):** Concept 1 + Concept 3 (Integrated Model)

**Reason for Shortlisting:**

**Combining AI-based image detection with offline questionnaire ensures both accuracy and accessibility under rural constraints. It balances feasibility, scalability, and farmer usability.**

**6.8 Ideation Summary (5 lines only)**

1.Multiple concepts were generated before selection.

2.Both AI and process-based solutions were evaluated.

3. Rural constraints influenced final direction.

4.Hybrid accessibility was considered essential.

5.Final shortlist balances accuracy and feasibility.

# Declaration:

☑ Journey mapping is based on real beneficiary interaction

☑ Constraints derived from field study

☑ Minimum 5 solution concepts developed

☑ No premature selection without comparison

ARUN KUMAR S

**Signature of the Student**